

GROUP 14

A GENERAL ANOMALY DETECTION FRAMEWORK FOR FLEET-BASED CONDITION MONITORING OF MACHINES

Kilian Hendrickx, Wannes Meert, Yves Mollet, Johan Gyselinck, Bram Cornelis, Konstantinos Gryllias, Jesse Davis

PROJECT FOCUS:

Early fault detection in machine fleets before failure occurs

KEY INNOVATION:

- Fleet-based approach leveraging similar machines
- Unsupervised anomaly detection without historical labeled data
- Real-time online monitoring with visualization

EGDE MEMBERS:

DWARAKESH VENKATESH - CB.SC.U4CSE23020
SHYAM SUNDAR RAJU - CB.SC.U4CSE23049
SARIGASINI M - CB.SC.U4CSE23065
POOJA SHREE S - CB.SC.U4CSE23346

CI MEMBERS:

ASWATH SIDDHARTH - CB.SC.U4CSE23109
DARSHAN R - CB.SC.U4CSE23115

THE EDGE LAYER

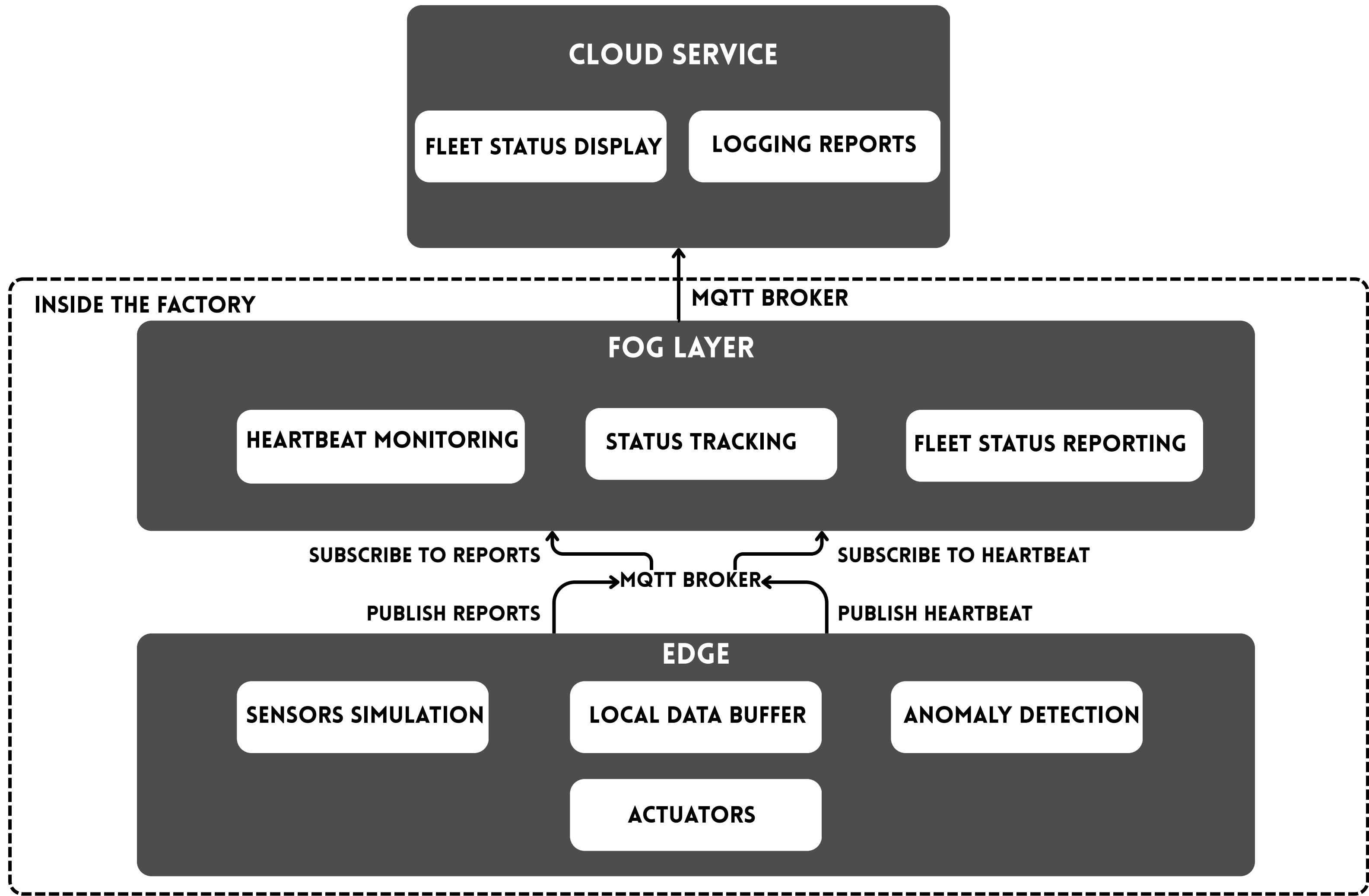
- **Sensor Simulation:** It generates realistic, time-varying data for multiple metrics like temperature, humidity, and pressure.
- **Local Data Buffering:** It maintains a rolling window (a local buffer) of the most recent sensor readings.
- **Edge-based Anomaly Detection:** It performs real-time anomaly detection locally on the data buffer, reducing the need to send all raw data to the cloud.
- **MQTT Communication:** Uses the Message Queuing Telemetry Transport protocol for machine-to-machine (M2M) communication
- **Publish Reports:** Sends detailed analysis reports (decision, anomaly score, metrics) to the cloud through fog for further processing and fine-tuning.
- **Publish Heartbeats:** Sends regular “alive” status messages to the fog-layer.
- **Actuator triggering:** Can trigger actuators on its own without instruction from cloud or fog.

THE FOG LAYER

A heartbeat is a small, periodic signal or message that an edge node sends to a central controller (or monitoring system) to indicate that it is still active and functioning.

- **Heartbeat Monitoring:** It subscribes to a wildcard MQTT topic (edge/+/heartbeat) [+ stands for any node ID] to listen for "alive" signals from all edge nodes. The payload of the heartbeat might be very simple, a string like “alive”.
- **Status Tracking:** It maintains the status of each edge node. If a node fails to send a heartbeat within a configurable HEARTBEAT_TIMEOUT, it's marked as "unresponsive". When a heartbeat resumes, it's marked as "alive" again.
- **Fleet Status Reporting:** It periodically compiles a summary report (a "fleet snapshot") detailing the status of every known edge node and publishes it to the cloud.

This mechanism essentially ensures that the edge devices are functional, and faults in the edge nodes are detected without delay.



THE CLOUD LAYER

- **MQTT Subscriber:** It connects to an MQTT broker and subscribes to two topics to gather data from the entire IoT system:
 - **Reports:** To receive anomaly analysis reports from each fog node.
 - **Fleet_snapshot:** To receive periodic health status summaries for the entire fleet.

This data is retrieved at the fog layer.

- **Responsive UI:** The dashboard automatically refreshes once data is received, providing a live view of the system. The data is received at periodic intervals, with the period being determined by the architect.
- **Fleet Snapshot:** Displays the overall fleet size and status. It shows a table listing each edge node, its current status, and its latest reported anomaly score. It also includes a bar chart visualizing the anomaly scores for all currently “alive” nodes.
- **Live Report Stream:** Shows a continuously updating table of detailed reports as they arrive from the edge nodes, updating periodically.

UNSUPERVISED LEARNING MODEL

Hierarchical Clustering: The framework applies agglomerative hierarchical clustering to group machines by pairwise similarity. A correlation threshold is used to cut the dendrogram into meaningful clusters, avoiding arbitrary distance cutoffs.

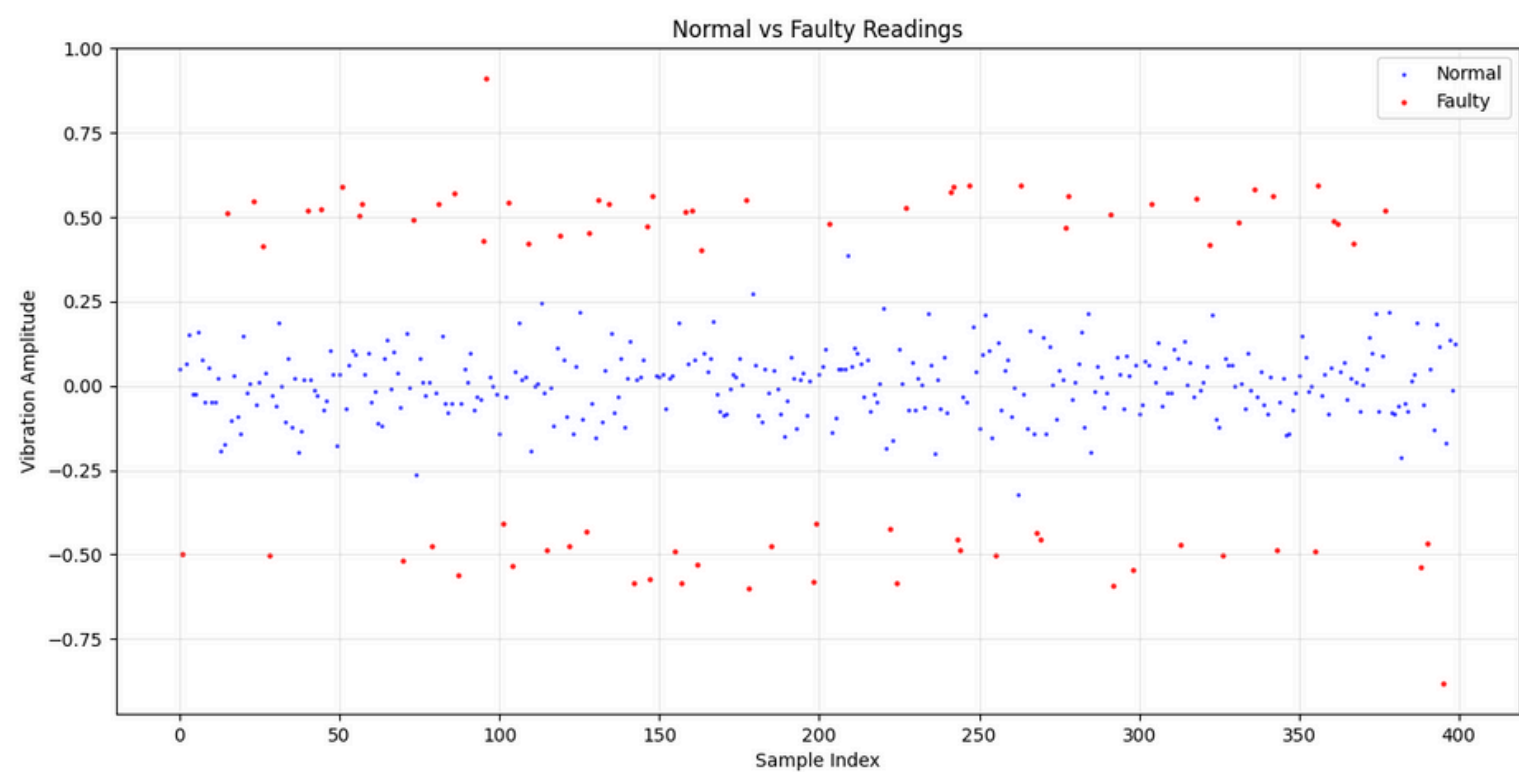
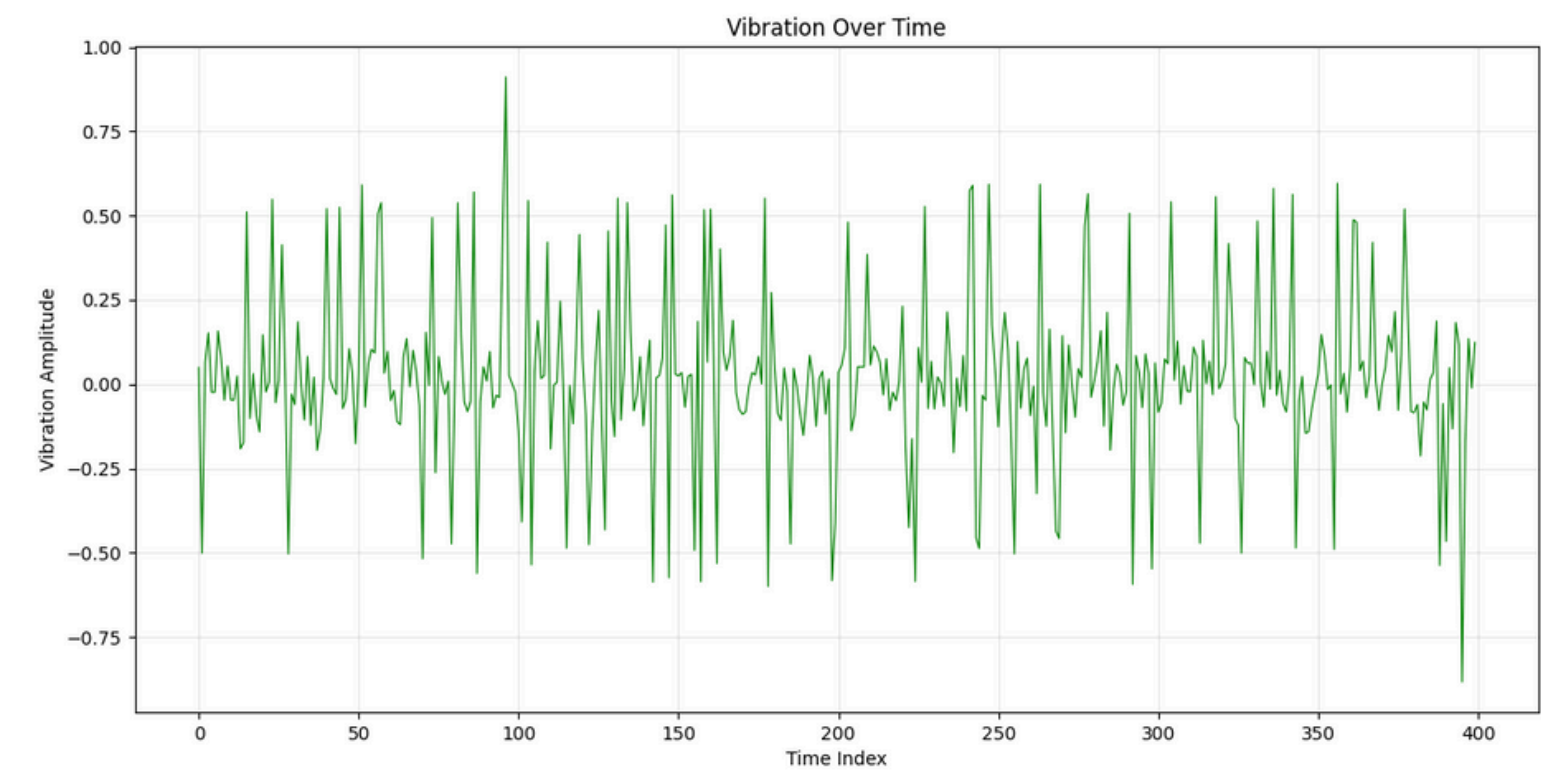
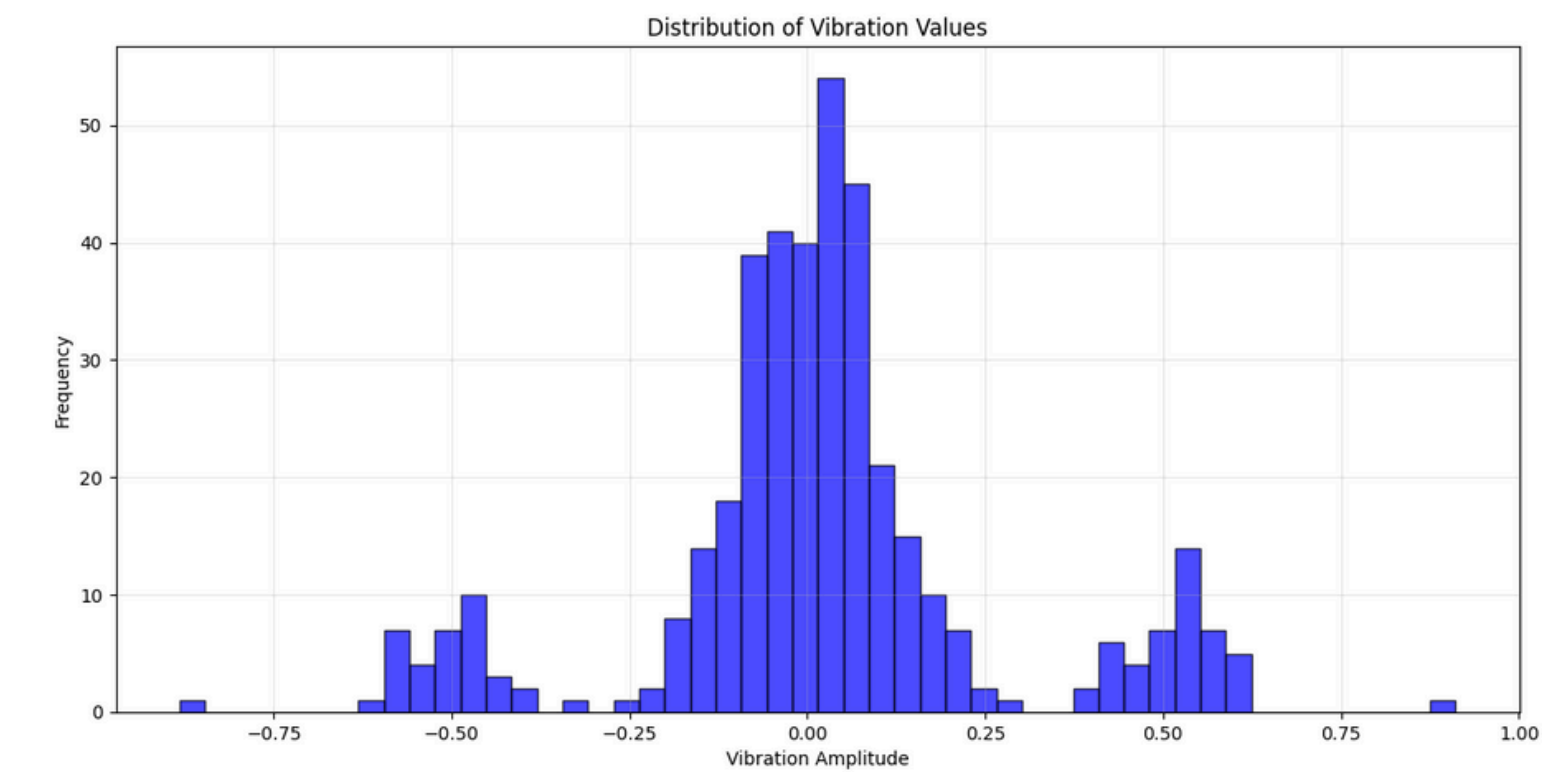
Anomaly Identification: Most machines are assumed healthy. Clusters representing the majority are considered normal, while smaller clusters are flagged as anomalous.

Anomaly Scoring: A simple fraction-based score is assigned to each machine: the proportion of the fleet not in its cluster. Scores close to zero indicate healthy behavior. A threshold (typically 2/3) determines the final classification. $1 - (\text{size of its cluster} / \text{total number of machines})$ is the formula used.

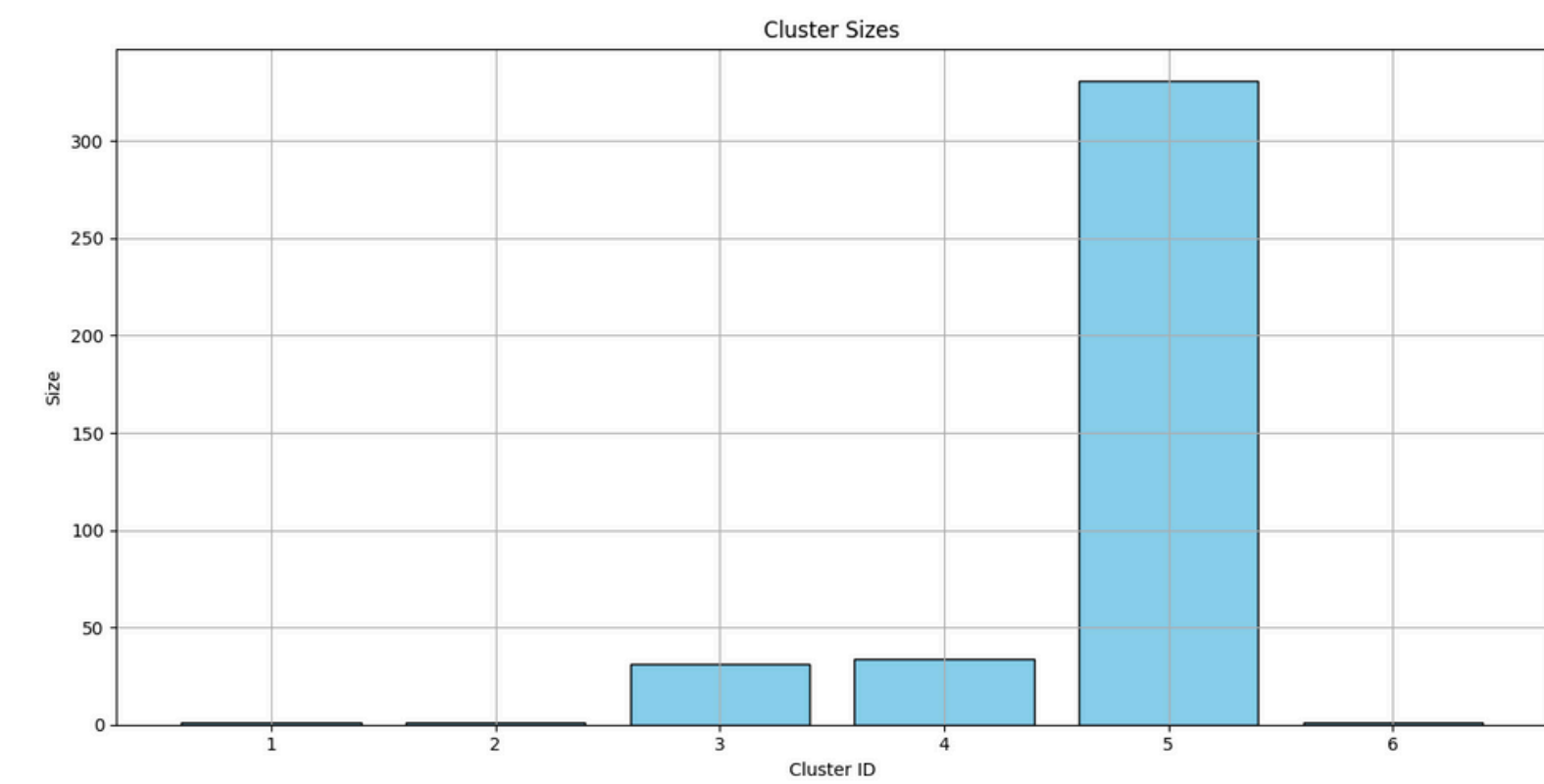
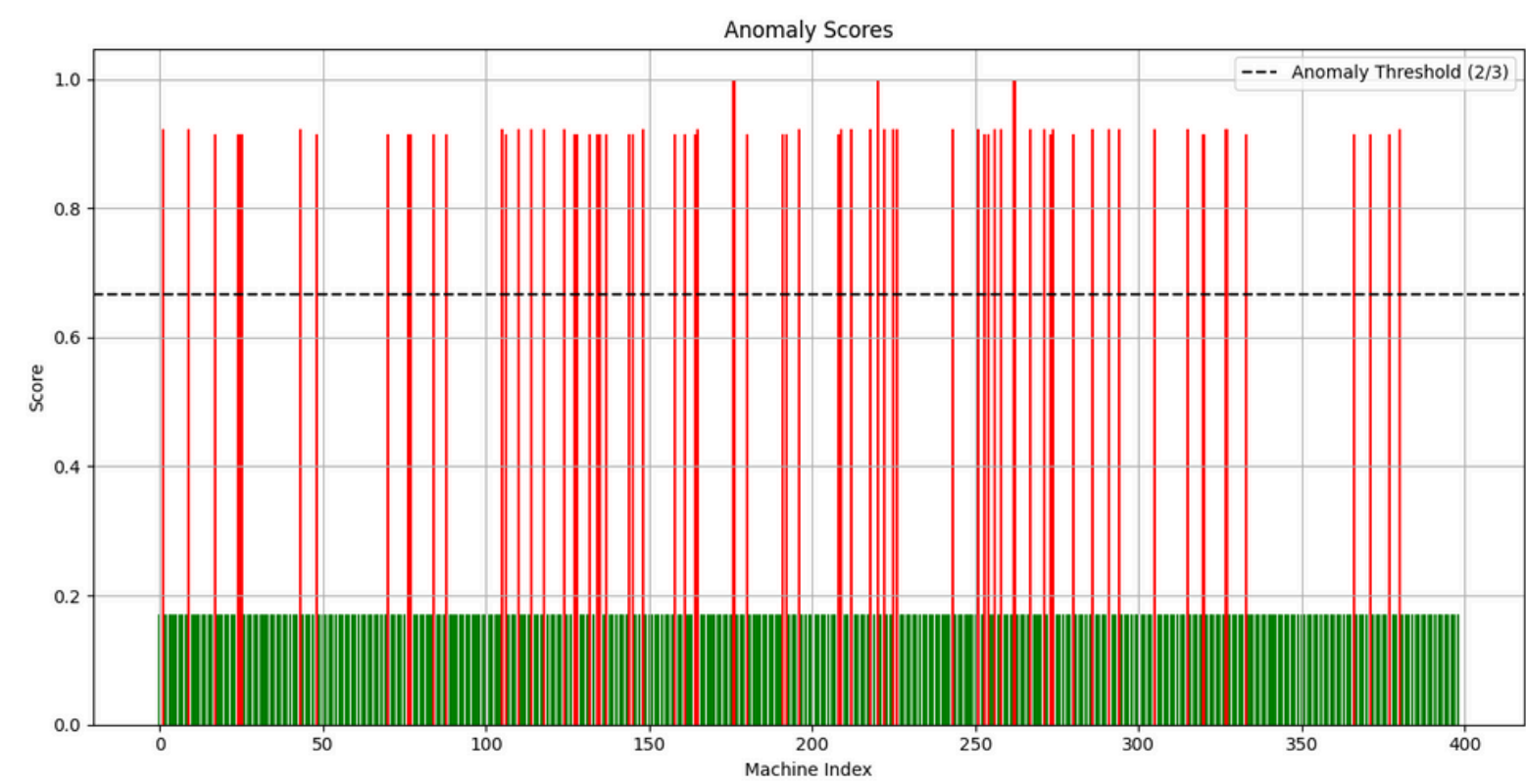
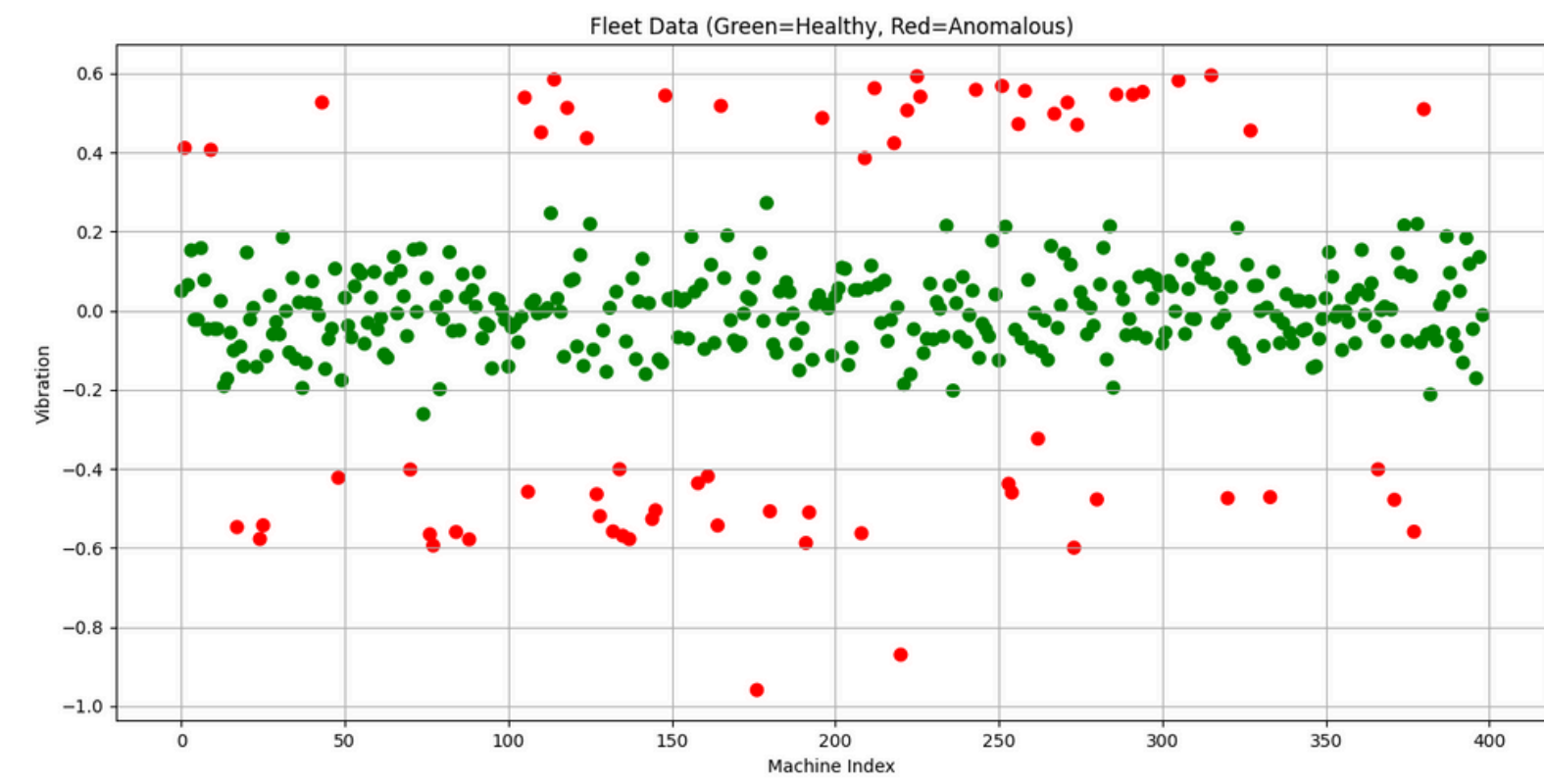
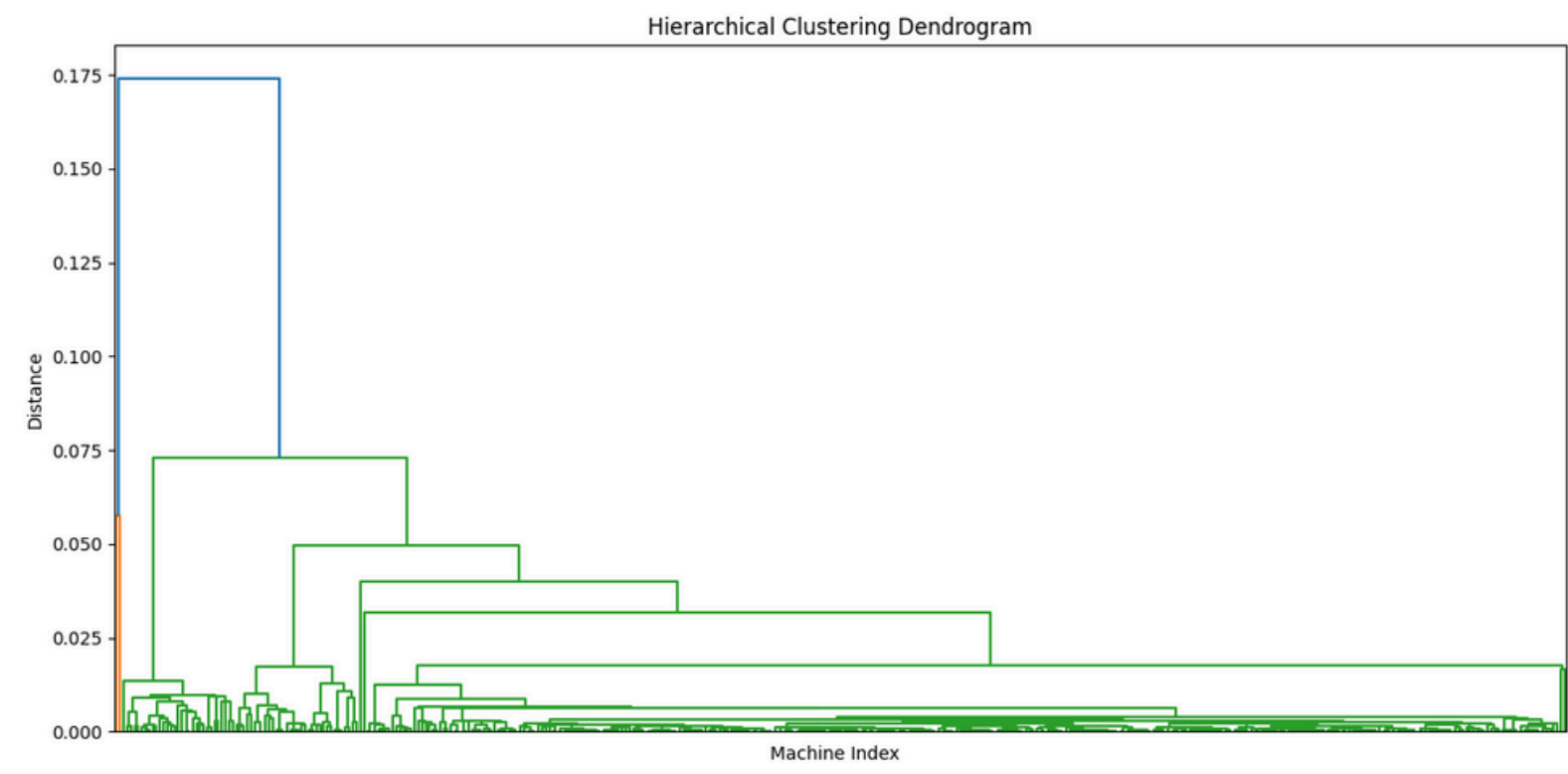
Visualization: The method emphasizes interpretability with plots such as normalized signals, dissimilarity matrices, dendrograms, and anomaly scores, visualizing reasoning behind predictions.

Benchmark Method: For evaluation, the framework is compared to a simpler statistical rule that labels machines anomalous when their measurements fall outside a few standard deviations from the fleet mean.

VISUALIZING THE DATASET



VISUALIZING THE MODEL PREDICTIONS



ENHANCING ANOMALY DETECTION: BEYOND RIGID RULES

- **Current Model Limitations:**
 - Hard Clustering: Machines are strictly assigned to one group (100% in/out), ignoring ambiguous states.
 - Problem: A machine trending towards failure is either "OK" or "Faulty," no middle ground.
 - Rigid Thresholds: Anomalies are flagged by fixed numerical cut-offs.
 - Brittle: Small data fluctuations near a threshold cause abrupt "OK" to "ALERT" shifts.
 - Lack of Nuance: Cannot differentiate between minor deviations and critical issues.
 - [cite_start]The original paper's DTW model, while effective, shows F1-score sensitivity to its hard thr_cc clustering threshold (e.g., for "820 RPM - load," F1 varies from 0.348 to 0.923 based on thr_cc selection)
- **Our Fuzzy Logic Enhancement:**
 - The Problem: Real-world machine behavior is nuanced and uncertain.
 - Our Solution: Integrated Fuzzy Logic to handle this inherent ambiguity.
 - Core Idea: Replaced rigid "black & white" decisions with continuous "shades of grey."

FUZZY LOGIC IN ACTION: THE MECHANISM

- How Our Fuzzy Model Provides Granular Insights:
 - Fuzzy C-Means (FCM):
 - Instead of hard assignments, machines are given degrees of membership to all clusters.
 - Example: Machine A can be 70% in a "Healthy" cluster and 30% in a "Deviant" cluster.
 - This directly models the uncertainty and partial belonging, capturing transitional behaviors.
 - Fuzzy Anomaly Score Calculation:
 - A continuous anomaly score is derived from these membership degrees:
 - Fuzzy Anomaly Score = $1 - (\text{Membership in the largest 'Healthy' Cluster})$
 - Higher score = greater likelihood of anomaly.
 - Fuzzy Inference System (FIS):
 - This is the "brain" that translates the Fuzzy Anomaly Score into actionable insights.
 - It uses expert-defined, intuitive IF-THEN rules to determine machine health.
 - Example Rules:
 - IF AnomalyScore IS Low THEN HealthStatus IS Healthy
 - IF AnomalyScore IS Medium THEN HealthStatus IS Warning
 - IF AnomalyScore IS High THEN HealthStatus IS Faulty
 - Output: A precise, continuous Health Score (0-100) for each machine.

IMPACT: PROACTIVE MAINTENANCE & PERFORMANCE CONTEXT

- Key Benefits of Our Fuzzy Logic Approach:
 - Proactive Maintenance: The detailed Health Score (0-100) enables multi-tier alerts:
 - Healthy | Warning | Faulty
 - The "Warning" state is crucial for early intervention, allowing maintenance to be scheduled before critical failure, preventing costly downtime.
 - Enhanced Robustness: More resilient to noisy sensor data and subtle, gradual changes in machine behavior.
 - Interpretable AI: The transparent, rule-based nature of the FIS builds trust and makes decisions understandable for operators and engineers.
- Performance Context (Original Method):
 - [Insert the image of the performance table here]
 - Caption: Example F1-scores for the original DTW-based model, illustrating its sensitivity to hard thresholds.
 - Goal: Our Fuzzy Model aims to achieve comparable or superior performance to the original model, but with significantly increased robustness, adaptability, and interpretability, especially in handling the ambiguous, "grey" areas of machine health.
 - This reduces reliance on rigid, manually-tuned thresholds, making the system more practical for diverse industrial environments.

ME-FEEL AT THE EDGE: THE MECHANISM

How Our Edge Model Participates in Federated Learning:

Multi-Exit Network Design:

- Each edge device hosts a neural network with multiple exits.
- Lightweight nodes (STM32) train shallow exits.
- More powerful nodes (RPi/IPC) train deeper exits.

Exit-Specific Training:

- Devices use their buffered sensor data to update only the layers they can handle.
- Example: Edge A trains Exit 1 (early layers), Edge B trains Exit 2, Edge C trains Exit 3.

Model Update Publishing:

- After local training, updated layers are serialized and sent via MQTT.
- Topic: edge/{id}/model_update.
- Payload includes Edge ID, round number, exit depth, and weight updates.

FOG AGGREGATION & SYNCHRONIZATION

How the Fog Builds a Global Model:

Layer-Wise Averaging:

- Fog collects all model updates from different exits.
- For each shared layer, weights are averaged across edges.
- Produces a unified, global multi-exit model.

Global Model Distribution:

- Aggregated model is broadcast via MQTT to fog/global_model.
- Edges download updates and synchronize their state dictionaries.

Example Flow:

1. Edge1 trains Exit 1 → sends shallow layers.
2. Edge2 trains Exit 2 → sends deeper layers.
3. Fog averages and publishes global model.
4. All edges update and continue training.

IMPACT OF ME-FEEL IN FLEET MONITORING

Key Benefits of Integrating ME-FEEL:

Adaptivity:

- Heterogeneous devices contribute without overload.

Efficiency:

- Only partial weights are exchanged → lower bandwidth and energy use.

Scalability:

- Larger fleets improve global model accuracy.

Resilience:

- Training continues even if weaker or unstable nodes drop out.

Enhanced Anomaly Detection:

- Improved feature extraction at the edge strengthens fleet-level clustering and anomaly scoring.

WHAT HAVE WE DONE TILL NOW?

- **Dataset collection and visualization:** This project makes use of the IMS Bearing Dataset, which is a well-known dataset in machine condition monitoring. Available at - <https://data.nasa.gov/dataset/ims-bearings>
- **Anomaly Identification through unsupervised learning:** Any machine that deviates too much from the majority “healthy” cluster is identified as an anomaly, and is appropriately identified based on a set threshold.
- **Anomaly Scoring:** An anomaly score is generated for each machine, and based on a set threshold, the machines are classified into clusters, and the number of clusters is determined automatically to differentiate between the level of fault in each machine
- **Visualization:** The machine metrics can be visualized, and the anomaly scores of each machines can be seen in addition to plotting the dendrogram so that there is complete transparency in the model, and any changes required can be seen clearly.



WHAT ARE WE WORKING ON?

- **Pushing the model to an edge device:** The accuracy of the model is high but it has to be optimized to work at the edge level - essentially a tradeoff in accuracy to keep up with the performance of the edge device.
- **Edge node limitations:** The CPU and RAM limitation are yet to be enforced through the docker container. It is still kept pending to monitor the working of the entire architecture.
- **Remove the development features:** The features added to debug the project should be removed once all other features are completed.
- **Integrate CI models into the edge layer and compare the results.**